KIRILL BRILLIANTOV

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EDUCATION

EPFL, MDS Lab — **Sep 2025 - Now**
Doctoral Program in Computer and Communication Sciences — *Lausanne, Switzerland*
Advised by Emmanuel Abbe and Samy Bengio

ETH Zürich (ETH) — **Sep 2023 - Jun 2025**
Master of Computer Science, GPA: **5.51 / 6.0** — *Zürich, Switzerland*

Combined Bachelor's GPA: 8.99/10.0 — **Bachelor's thesis:** [4]

Constructor (Jacobs) University Bremen (C(J)UB) — **Sep 2022 - Jun 2023**
completed Bachelor of Computer Science, GPA **1.5 / 1.0** — *Bremen, Germany*

Higher School of Economics Saint-Petersburg (HSE SPb) — **Sep 2019 - Aug 2022**
discontinued Bachelor of Computer Science, GPA **9.08 / 10.0** — *Saint-Petersburg, Russia*

(selected) RESEARCH EXPERIENCE

Autonomous Machine Learning Engineering, [1] — **Mar 2026 - Sep 2026**

Extensively benchmarked OS models and harnesses on MLE tasks; a plain coding agent performed no worse
Keywords: evaluation, agents, machine learning engineering, statistical analysis

Applying AI to k-server conjecture, [2] — **Sep 2025 - Jan 2026**

Reformulated a proof technique as an optimization problem and applied AlphaEvolve-style methods to improve SotA
Keywords: AI for Science, agents, AlphaEvolve, evaluations

Generalization of Persistent Homology (PH), [3] — **Jun 2023 - Aug 2023**
research internship at Aalto University under supervision of Vikas Garg, Amauri Souza

Developed a **compositional** PAC-Bayes framework and generalization bounds for PH and PH-augmented GNNs
Keywords: GNNs, PH, PAC-Bayes, generalization, theory

Applying Language Models to Algebraic Topology, [4] — **Feb 2022 - May 2023**
bachelor's thesis at JUB, HSE SPb under supervision of Fedor Pavutnitksiy

Used a combinatorial reformulation of $\pi_n(S^3)$ and trained transformers on free group subsets to find their intersection
Keywords: Sphere Homotopy, SFT, synthetic data, AI for Science, LM training

$(n, 4)$-MaxSAT and General MaxSAT, [7] — **Sep 2021 - Jan 2022**
coursework at HSE SPb under supervision of Ivan Bliznets

Developed theoretically faster algorithms for solving $(n, 4)$-MaxSAT and general MaxSAT instances
Keywords: theory, complexity analysis, MaxSAT, NP problems, branch and bound

WORK EXPERIENCE

Research Scientist — **Apr 2026 - Now**
contractor at Apple, Machine Learning Research (MLR) — *Lausanne, Switzerland*

Part of a research group led by Emmanuel Abbe, working on autonomous AI agents

Research Assistant — **Apr 2024 - Aug 2025**
part-time job at Strategic Management and Innovation lab (ETH Zürich) — *Zürich, Switzerland*

Supported social science research by scraping data and building models and visualizations. *Example project*: Contributions of chip manufacturers to open-source ML frameworks

Research Project — **Sep 2024 - Jan 2025**
supervised by Sergei Gukov (Caltech) — *Remote*

Explored machine learning and reinforcement learning approaches to computing Kazhdan's property (T)

Research Assistant — **Oct 2023 - Aug 2025**
several projects during masters at LAS Groups (ETH Zürich) — *Zürich, Switzerland*

Optimal transport: Charlotte Bunne; RL theory: Marco Bagatella; LLM safety: Xin Chen and Marin Vlastelica

Research Intern **Jun 2024 - Aug 2024**
internship at Robinson Group (ISTA) *Vienna, Austria*

Benchmarked summary statistics methods against classical methods for inferring genetic effects

Research Intern **Jun 2023 - Aug 2023**
internship at QuML (Aalto University) *Helsinki, Finland*

Research Assistant **Feb 2022 - Jul 2022**
part-time job at EIMI *Saint Petersburg, Russia*

Intern Backend Engineer **Jul 2021 - Oct 2021**
internship at Yandex.Direct *Saint-Petersburg, Russia*

Part of a team maintaining the Yandex.Direct API responsible for showing ads on Yandex.Browser

TEACHING AND ACADEMIC ACTIVITIES

AI for Science European Summer School **Jul 2026**
Hellenic Institute of Advanced Studies; sponsored by Google DeepMind *Athens, Greece*

Reviewer **2025–2026**
NeurIPS 2025, ICLR 2026, NeurIPS 2026 Datasets and Benchmarks Track

Teaching Assistant. Probabilités et statistique (pour IC) **Sep 2026 - Jan 2027**
EPFL *Lausanne, Switzerland*

Teaching Assistant. Introduction to ML **Apr 2024 - Jun 2024**
paid position at ETH Zürich *Zürich, Switzerland*

Helped typeset the lecture script; wrote one chapter (≈ 3 hours of lectures) with handmade visualizations

Teaching Assistant. C++ **Sep 2022 - Dec 2022**
volunteer at HSE SPb *Remote*

Reviewed home assignments of **20** freshmen. Gave feedback on readability, style, architecture, and correctness.

Teaching Assistant. Mathematical Logic **Apr 2022 - Jul 2022**
paid position at HSE SPb *Saint-Petersburg, Russia*

Checked **homework** and taught **tutorials** for **14** freshmen, covering equinumerosity of sets, Boolean functions and circuits, and basics of proof theory

Coursework Supervisor. C++ **Feb 2022 - Jun 2022**
volunteer at HSE SPb *Saint-Petersburg, Russia*

Mentored a group of **3** freshmen, helping solve technical and architectural problems, organizing regular calls, and reviewing code. The commission **highly** rated their result

PUBLICATIONS (by recency)

[1] **Kirill Brilliantov**, Alejandro Hernández-Cano, and Emmanuel Abbé. *How Much of a Harness Does a Strong Agent Need for Autonomous ML Engineering?* 2026. arXiv: 2609.40303 [cs.AI]. URL: https://arxiv.org/abs/2609.40303.

[2] [workshop] **Kirill Brilliantov**, Etienne Bamas, and Emmanuel Abbé. *k-server-bench: Automating Potential Discovery for the k-Server Conjecture*. 2026. arXiv: 2604.07240 [cs.MS]. URL: https://arxiv.org/abs/2604.07240.

[3] [NIPS] **Kirill Brilliantov**, Amauri H. Souza, and Vikas Garg. "Compositional PAC-Bayes: Generalization of GNNs with persistence and beyond". In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. 2024. URL: https://openreview.net/forum?id=ZNcJtNN3e8.

[4] [ICML] **Kirill Brilliantov**, Fedor Pavutnitskiy, Dmitry Pasechnyuk, and German Magai. "Applying language models to algebraic topology: generating simplicial cycles using multi-labeling in Wu's formula". In: *Forty-first International Conference on Machine Learning*. 2024. URL: https://openreview.net/forum?id=gQz30hTkRE.

[5] [workshop] Denis Tarasov, **Kirill Brilliantov**, and Dmitrii Kharlapenko. "Is Value Functions Estimation with Classification Plug-and-play for Offline Reinforcement Learning?" In: *Automated Reinforcement Learning: Exploring Meta-Learning, AutoML, and LLMs*. 2024. URL: https://openreview.net/forum?id=UoVfuqUOtD.

[6] [AAAI] Vasily Alferov, Ivan Bliznets, and **Kirill Brilliantov**. "Parameterization of (Partial) Maximum Satisfiability above matching in a variable-clause graph". In: *Proc. Conf. AAAI Artif. Intell.* 38.8 (Mar. 2024), pp. 7918–7925.

[7] [AAAI] **Kirill Brilliantov**, Vasily Alferov, and Ivan Bliznets. "Improved algorithms for Maximum Satisfiability and its special cases". In: *Proc. Conf. AAAI Artif. Intell.* 37.4 (June 2023), pp. 3898–3905.